

Math 231. Practice Exam I. Fall 2022

Full Name: _____

Section Code:

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Net ID: _____

Version A

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- **Do not turn this page until instructed to!**
 - You must not communicate with other students during this test. No written materials allowed.
 - No phones, calculators, smart-watches or electronic devices of any kind allowed for any reason.
 - Violations of academic integrity (in other words, cheating) will be taken extremely seriously.
 - Point values are indicated next to each question.
 - You have **50 minutes**. Good luck!

Did you fill out your name? —

Did you fill out your Net ID? —

Did you fill out your Section Code? —

Trig identities: $\sin^2 x + \cos^2 x = 1$, $\tan^2 x + 1 = \sec^2 x$, $\sin 2x = 2 \sin x \cos x$,
 $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

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Problem 1

Determine whether the following statements are True or False. **NO NEED TO JUSTIFY.**

(a)

$$\int \frac{1}{x^2 + 1} dx = \int \frac{1}{u} du = \ln |u| + C = \ln |x^2 + 1| + C.$$

(b)

The polynomial $x^2 + x + 1$ is irreducible.

(c)

The polynomial $x^3 + 1$ is irreducible.

(d)

$$\int \frac{x}{x+1} dx = x - \int \frac{1}{x+1} dx.$$

Problem 2

For the following problems, just find the partial fraction decomposition and solve for the constants. **DO NOT INTEGRATE**

(a)

$$\frac{3x}{(2x+1)(x-1)}$$

(b)

$$\frac{x-4}{4x^2-20x+25}$$

(c)

$$\frac{3x^2+1}{x^3-2x^2+x}$$

Problem 3

Evaluate the following integrals. Show all your work and box your final answer.

(a)

$$\int 5 \cos^4(x) \, dx$$

(b)

$$\int \frac{2e^x}{e^{2x} + 8e^x + 12} \, dx$$

(c)

$$\int \frac{7x^4}{\sqrt{(x^2+1)^7}} \, dx$$

(d)

$$\int \tan^{-1}(t) \, dt$$

Problem 4

For each of the four integrals below, just state which of the following methods applies: u -substitution, integration by parts, trigonometric substitution, or partial fractions.

You do not need to justify your answer. **DO NOT INTEGRATE.**

(a)

$$\int x^2 \sec^2(x^3) \, dx$$

(b)

$$\int \theta^2 \ln(\theta) \, d\theta$$

(c)

$$\int \frac{-4t + 5}{t^4 + t^3 + t^2} \, dt$$

(d)

$$\int \frac{x^2}{(x^2 - 4)^{3/2}} \, dx$$